UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Graphs and diagrams

Every graph and diagram in scope, redrawn as clean vector figures.

Class 003 · Particular integral Type 1 ($X = e^{ax}$) · compiled 15 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S003·8	Unit 2 deck 003 (“Particular integral of Type 1”), slide 8 (rendered .pptx)
T003 00:10:15	Unit 2 AV-summary transcript 003 (“Particular integral of standard functions – Type 1”, 21 min), segment start
T004a 00:02:06	Unit 2 AV-summary transcript in bucket 004 (“Particular integral Type 1 and Type 2”, 12.6 min); only its Type 1 part (00:00:00–00:06:04) and the constant term of its $\sin^2 x$ example are cited here. The class 004 pack cites the same file.
CW U2 P10	Unit 2 class-work problem 10
ESA Jan-24 s1 2d	Past paper: UE23MA141A Jan 2024 set 1, question 2.d

Contents

1. Figure G3.1 · The failure ladder: the PI divided by e^x
2. Figure G3.2 · Forcing e^{-x}: e^{-x}, xe^{-x} and x^2e^{-x} particular integrals
3. Figure G4.1 · $\sinh x$ and $\cosh x$ are sums of two exponentials
4. Figure G4.2 · $a^x = e^{x \log a}$: 2^x and 2^{-x}

The deck has no graphs, so every figure is added for clarity. Colours follow the Okabe–Ito order: blue = no failure, vermilion (dashed) = one failure, green = two failures; grey dashed lines are references. Every plotted particular integral was found by the Type 1 rule and checked by substituting it back with sympy.

3 Type 1: $X = e^{ax}$ and the cases of failure

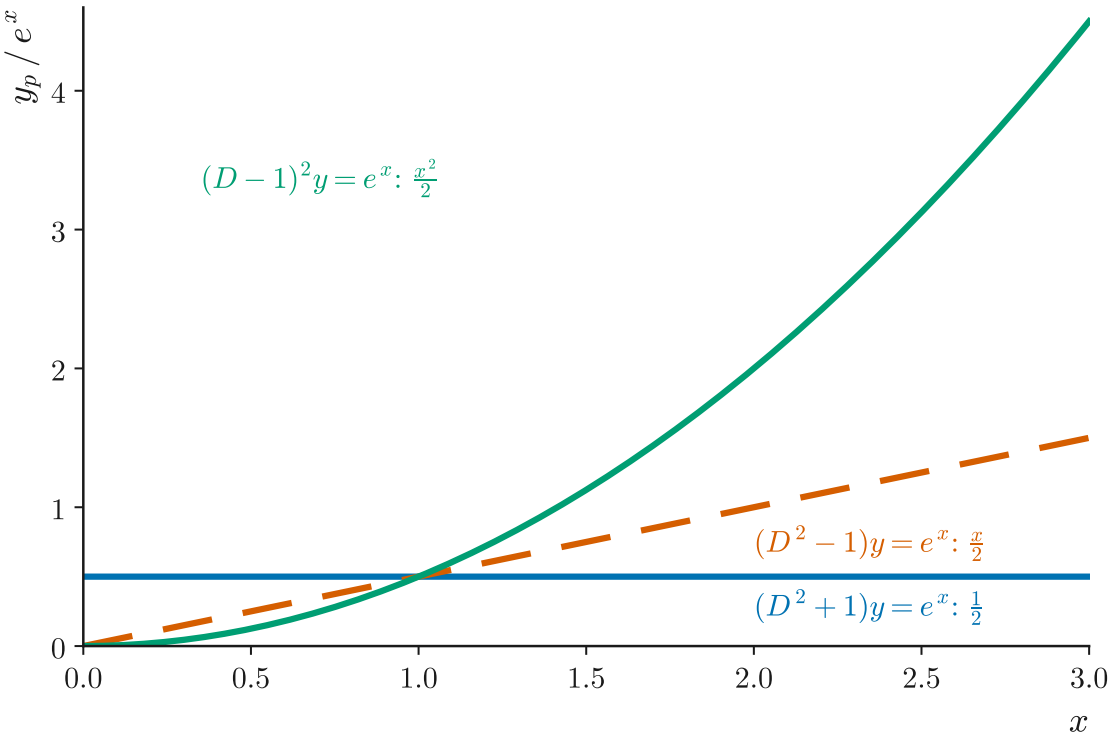


Figure G3.1. The failure ladder: the PI divided by e^x. ESA ★★★

ADDED FOR CLARITY S003·8 · S003·13 · T003 00:12:39

How to read it. Three equations with the same forcing e^x ($a = 1$). If 1 is not a root ($D^2 + 1$, $f(1) = 2$) the PI is a constant multiple $\frac{1}{2}e^x$ (blue, flat ratio). If 1 is a simple root ($D^2 - 1$, $f'(1) = 2$) the ratio grows like $\frac{x}{2}$ (vermilion dashed). If 1 is a double root ($(D-1)^2$, $f''(1) = 2$) it grows like $\frac{x^2}{2}$ (green). Each failure adds one power of x, a resonance-like growth.

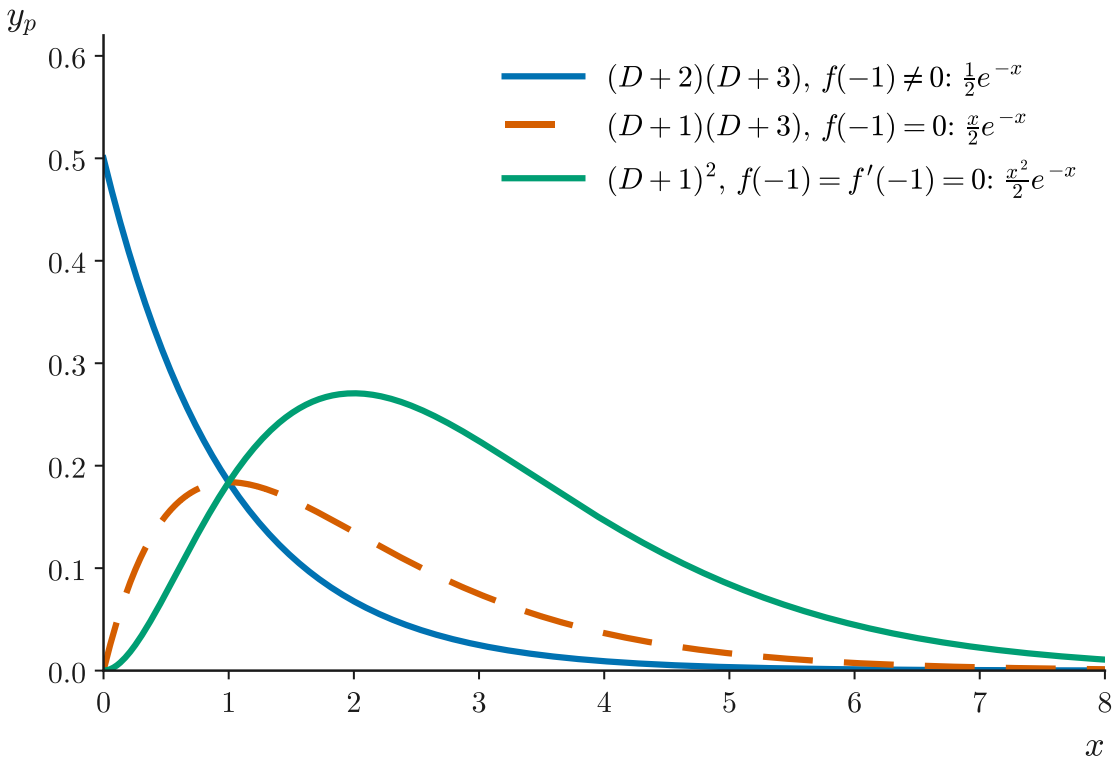


Figure G3.2. Forcing e^{-x}: e^{-x}, xe^{-x} and x^2e^{-x} particular integrals. ESA ★★

ADDED FOR CLARITY S003·8 · S003·15

How to read it. For $(D+2)(D+3)$, $(D+1)(D+3)$ and $(D+1)^2$ with $X=e^{-x}$ the PIs are $\frac{1}{2}e^{-x}$ (blue), $\frac{x}{2}e^{-x}$ (vermilion dashed) and $\frac{x^2}{2}e^{-x}$ (green). Without failure the PI just copies the shape of X; with a failure it rises before it decays, because e^{-x} is already in the CF. Compare slide 003·15, $\frac{x^2}{2}e^{-2x}$.

4 Extensions: constants, $\sinh ax$, $\cosh ax$ and a^x

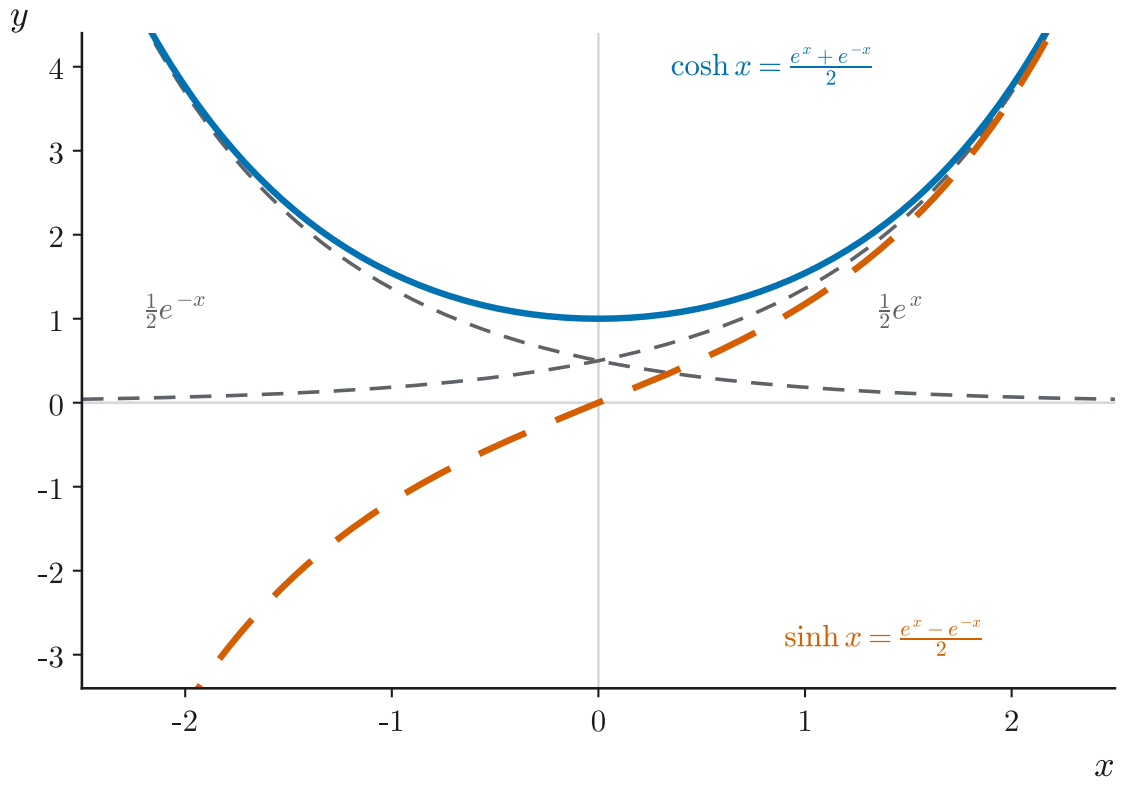


Figure G4.1. $\sinh x$ and $\cosh x$ are sums of two exponentials. ESA ★★

ADDED FOR CLARITY S003·9 · S003·10

How to read it. $\cosh x$ (blue) is the sum and $\sinh x$ (vermilion dashed) the difference of $\frac{1}{2}e^{x}$ and $\frac{1}{2}e^{-x}$ (grey). So $\frac{1}{f(D)}$ acts on each half separately, with $D \to a$ and $D \to -a$ (E4.2, E4.3).

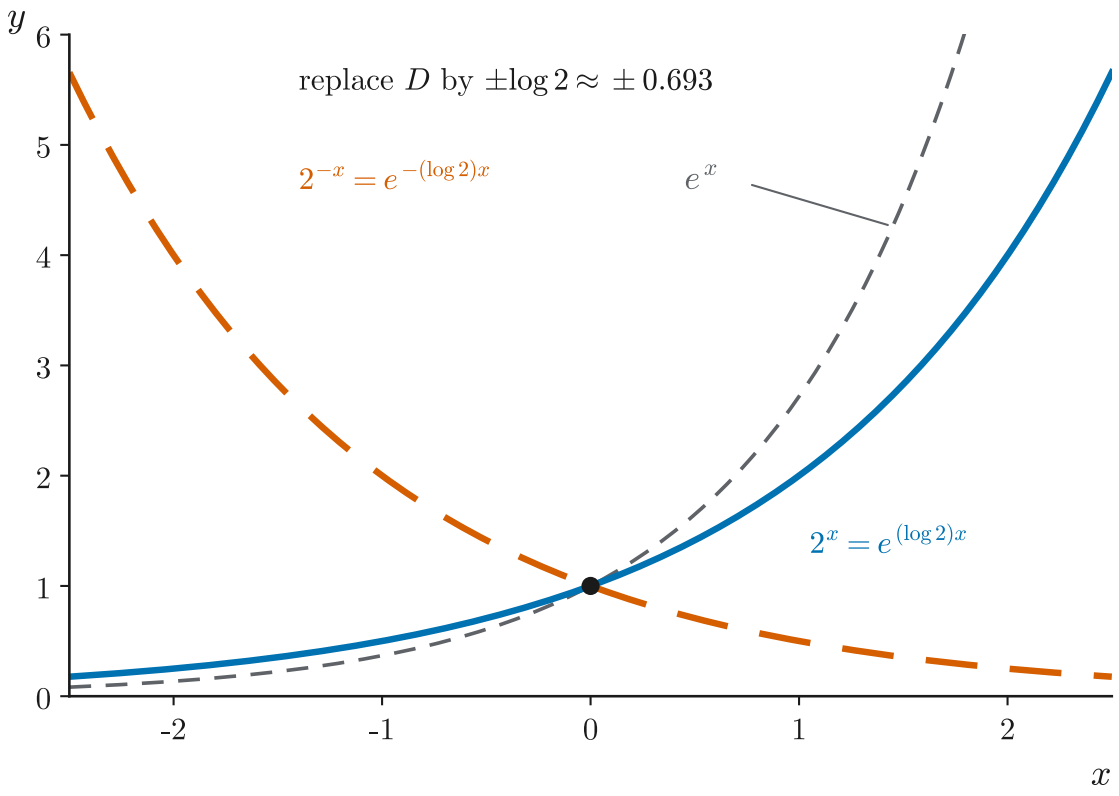


Figure G4.2. $a^x = e^{x \log a}$: 2^x and 2^{-x}. ESA ★★

ADDED FOR CLARITY S003·10 · T003 00:16:11 · ESA Jan-24 s1 2d

How to read it. 2^x (blue) is an exponential with exponent $\log 2 \approx 0.693$, flatter than e^x (grey). 2^{-x} (vermilion dashed) has exponent $-\log 2$. All pass through $(0, 1)$. In the Type 1 rule, D is replaced by the exponent: $\pm \log 2$, never by 2.